School of Computer Science

Applied Machine Learning Project

# Predicting the Results of Professional Tennis Matches with Machine Learning


Author: Sergio de Carvalho, D24130831


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# Declaration

I hereby declare that the work described in this report is, except where otherwise stated, entirely my own and that I have not submitted it before for summative assessment in third-level education. I have read and understand TU Dublin's policy on plagiarism.

Signed ....... Joe Bloggs .......          Date ...... 02/04/2026 ......

(Sergio de Carvalho, D24130831)

## Acknowledgements

[Please leave this section out if you do not have any acknowledgements.]

# Abstract

[This should be up to 200 words and provide a summary of what was done and important findings.]

# Contents

# Acronyms and Abbreviations

| | |
|---|---|
| **ANP** | Analytic Network Process |
| **ATP** | Association of Tennis Professionals |
| **DT** | Decision Tree |
| **EDA** | Exploratory Data Analysis |
| **GBM** | Gradient Boosting Machine |
| **LASSO** | Least Absolute Shrinkage and Selection Operator |
| **LGBM** | Light Gradient Boosting Machine |
| **ML** | Machine Learning |
| **MLP** | Multi-Layer Perceptron |
| **RF** | Random Forest |
| **SVM** | Support Vector Machine |
| **WTA** | Women's Tennis Association |
| **XGBoost** | eXtreme Gradient Boosting |

# 1 Introduction

Tennis is a popular sport where the one-on-one nature of competition provides an ideal environment for sports analytics, as match outcomes are determined by a more controlled set of variables than in team-based sports. While professional tennis has benefited from rigorous record-keeping and a wealth of statistics about every aspect of the game, accurately predicting match results remains a complex challenge due to the intricate interactions between player skills, form, physical and psychological state, as well as environmental factors such as the weather, court surface type, and even tennis ball characteristics that can significantly influence the performance of players and the outcome of a match.

## 1.1 Project description

The focus of this project is to develop a framework that prioritises the curation of high-quality data and the engineering of domain-driven features that can capture the multiple facets of the game and be used to predict the results of professional tennis matches. Central to this approach is the use of "enhanced covariates"—variables that are not directly observed but are derived through mathematical transformations or separate statistical models.

There has been significant research on developing statistical models to rank tennis players as alternatives to the official rankings published by the professional tennis organisations. In parallel, several studies have applied machine learning to predict the results of tennis matches, however, most of these studies were restricted to specific tournaments (e.g. Grand Slam tournaments only) or did not take full advantage of the available data and statistical models. This project aims to address these limitations by tapping into the vast amount of data available and using machine learning techniques to explore the predictive power of different statistical models proposed in the literature while also discovering new ways to improve the predictions of these models.

## 1.2 Project motivation and justification

Predicting the outcomes of sporting events has been a long-standing problem in the field of sports analytics, and can have significant implications for the development of training methods and strategies to improve the performance of professional athletes. It is also important for broadcasters and the media in general as it helps them to transform a passive viewing experience into an interactive, high-stakes narrative that drives viewer retention and fan engagement. Finally, it can be argued that predicting sporting events can help develop valuable models and gain insights that could be useful in diverse areas of human activity, from financial markets, to politics and international conflicts (Gu and Saaty 2019).

## 1.3 Project scope

This project is focused on predicting the results of professional tennis matches on the main events organised by the Association of Tennis Professionals (ATP), the men's tennis circuit. While there is a wealth of data available from the Women's Tennis Association (WTA) as well, limiting the scope to one of the two circuits allows for a more focused and comprehensive analysis in the limited time available for this project. We note that the focus of this project is modelling predictions based solely on pre-match data rather than using data collected while the match is in progress.

Finally, we note that there is significant interest in the literature in using betting odds to either improve predictions or compare their accuracy. While odds that are carefully curated and published by professional bookmakers can be a valuable source of information, this project does not use odds data nor does it focus on this aspect of the problem.

## 1.4 Report layout

[A sentence on each of the following sections, introducing the reader to the report.]

# 2 Literature review

The Association of Tennis Professionals (ATP) manages the men's circuit and maintains the official player ranking that is used to determine tournament eligibility, entry qualification, and player seeding. This ranking is based on a cumulative rolling points system. Players earn points for each match won in an official tournament, and points remain on a player's record for 52 weeks before expiring. The number of points awarded for each win is determined by the event category and the round in which the match is played (Williams et al. 2021). For instance, the champion of any of the four annual Grand Slam events (Australian Open, Roland Garros, Wimbledon, and the US Open) receives 2,000 points, whereas the winner of a first-round match in the same tournament who does not advance any further receives 50 points. In contrast, the champion of an ATP 250 event receives only 250 points.

While official rankings are a good indication of the current strengths and abilities of tennis players, and can generally be used to predict the winner of a professional tennis match, they obviously do not capture the full story and are far from being a reliable prediction tool, particularly when the difference in rankings between players is small.

## 2.1 Paired comparisons and rating mechanisms

Noting that official systems often reward frequent participation over absolute performance, researchers set out to develop alternative rating mechanisms that could capture players' abilities by looking at their past match results, particularly taking into account their opponents' strengths rather than the prestige of the tournament or the round in which the match is played.

One of the earliest attempts to develop an improved rating mechanism for tennis was proposed by Glickman (1999), who devised a dynamic paired comparison model to estimate the abilities of players over time. He showed that, given a particular choice of constants, his method is equivalent to the popular Elo rating system developed for chess players by Arpad Elo in 1978. While Glickman used match results to produce an alternative ranking for tennis players, he did not test its forecasting capabilities.

McHale and Morton (2011) pioneered a similar rating mechanism by building a Bradley-Terry type model (a popular model for handling paired comparisons) that utilised players' past match results using an exponential decay function to account for players' recent form (i.e. giving more importance to recent results than older ones). Their model also accounted for the score of the matches. For instance, a player who wins a match with a score of 6-0 6-0 is rewarded more than a player who wins with a score of 7-6 7-6 (presumably, a much tighter match). They also note that tennis is played on different surfaces (e.g. clay, grass, and hard courts) that can significantly change the performance of players, thus they also made further use of weighting techniques so that matches played on surfaces other than that of the tournament being considered are given less importance. They showed that their model provided superior forecasts to the official rankings using data from nine seasons of the ATP tour from 2000-2008, and could provide higher returns against bookmaker odds using a simple betting strategy.

Expanding on this relational logic, Knottenbelt, Spanias, and Madurska (2012) proposed a common-opponent stochastic model that relied on the transitive nature of tennis, predicting match outcomes by calculating the difference in service points won against opponents that both players had previously faced. They argued that the intuition behind this approach is that tennis is a transitive sport to a certain extent (i.e. if player A is better than player B and player B is better than player C, then, usually, player A is better than player C). However, as many avid tennis fans will attest, this is often not the case. Using this approach, they were nevertheless able to generate 3.8% long-term profit against the best odds offered by bookmakers for a set of over 2,000 tennis matches from the ATP and WTA tours of 2011.

Kovalchik (2016) tested the predictive performance of 11 published forecasting models for predicting the outcomes of 2,395 singles matches during the 2014 season. They found that an Elo-based model was the best performing model, achieving an accuracy of up to 70%, although this was still worse than a Bookmakers consensus model built by aggregating the odds offered by established bookmakers for each match, which achieved accuracy of 72%.

Subsequent research heavily explored the Elo rating system. Williams et al. (2021) compared standard Elo, surface-specific Elo ratings, and weighted composites against betting odds, demonstrating that Elo-based approaches are effective, offering comparable, and sometimes superior, forecasting performance to betting odds on Grand Slam matches between 2018 and 2020. Angelini, Candila, and De Angelis (2022) introduced the concept of a Weighted Elo (WElo), which incorporates the margin of victory (i.e., the number of games or sets won) into the standard Elo updating formula to better reflect players' strengths. They tested their model on a set of matches from the ATP and WTA tours between 2012 and 2020, and showed that the WElo was superior to the standard Elo approach as well as the Bradley-Terry type model of McHale and Morton (2011), among other models.

Fayomi et al. (2022) revisited the Bradley-Terry model, using it to compute win probabilities and new player ratings that demonstrated the profound impact of court surface (especially, clay vs. hard court) on a player's true ability. They applied the model to data from the ATP tour between January 2019 to September 2020, achieving similar results to that of McHale and Morton (2011).

## 2.2 Machine learning approaches

Predictive techniques transitioned from statistical models to complex machine learning (ML) approaches. An early example is Somboonphokkaphan, Phimoltares, and Lursinsap (2009), who deployed a Multi-Layer Perceptron (MLP) – a foundational class of fully-connected feedforward artificial neural network – that incorporated time-series history, environmental data in the form of surface type, and several statistical features such as winning percentage of first serves and total points won. They tested their model on a set of matches between 2003 and 2008 and reported accuracy rates ranging from 69% to 81%, although it should be noted that these results were restricted to a small set of Grand Slam matches only.

Recognising that historical data alone misses intangible factors, Gu and Saaty (2019) built an Analytic Network Process (ANP) – an advanced, non-linear decision-making framework – combining data analysis with expert human judgments, capturing unquantifiable variables such as a player's psychology, brainpower, and stamina. They reported that their model achieved accuracy of up to 85.1%, however, once again, this was restricted to a set of only 94 matches from the 2015 US Open tournament. Moreover, this approach required the collection of judgements from tennis experts based on their knowledge and expertise about the matches and the players, an approach that is clearly difficult to scale.

Gao and Kowalczyk (2021) built three machine learning models (Logistic Regression, Random Forest, and Support Vector Machine), performed an in-depth feature selection process, and compiled one of the largest tennis statistics datasets publicly available. They found that a Random Forest classifier achieved the best performance, achieving prediction accuracies exceeding 80%. They also identified serve strength—specifically the percentage of points won on first and second serves—as a key predictor of match outcomes. Wilkens (2021) conducted a comprehensive comparative study using various ML methods (Logistic Regression, Neural Network, Random Forest, Gradient Boosting Machine, and Support Vector Machine) on a dataset of over 39,000 matches from the ATP and WTA tours, and noted that differences in prediction performance among the various ML approaches were small.

Yue et al. (2022) applied exploratory data analysis (EDA) to examine variables related to match results and introduced new variables using the Glicko rating system of Glickman (1999), including surface-specific ratings, which were then used to build several statistical and machine learning models (Logistic Regression, Support Vector Machine, Neural Network, and Light Gradient Boosting Machine). The models were trained on a dataset of Grand Slam matches between 2000 and 2019, and the results showed only marginal differences in performance between the various models, with the best results achieved using surface-specific Glicko ratings, with up to 73.8% accuracy on the ATP matches.

Buhamra, Groll, and Gerharz (2025) compared several models including regression-based (logistic, LASSO, splines) and machine learning approaches (Classification Tree, Random Forest, linear and non-linear Support Vector Machine, Artificial Neural Network, and XGBoost) using Grand Slam matches from 2011 to 2021. Different features were considered including ATP ranking and points, Elo ratings, bookmaker odds, and age variables that take into account the presumably optimal age of a tennis player being between 28 and 32 years old. Moreover, they investigated different forecasting strategies including “expanding window” and “rolling window”. They found that XGBoost achieved the highest classification rates, with up to 83% accuracy with the expanding window strategy.

## 2.3 Identified research needs

Taken together, the studies reviewed in this section illustrate a recurring limitation: conclusions are often drawn from short calendar windows or focus almost exclusively on Grand Slam tournaments. Moreover, as noted by Wilkens (2021), the heterogeneous setup and data used throughout the literature also demand prudence when comparing or even generalising their findings. These shortcomings motivate the emphasis for the present work: assembling and curating a large, multi-decade body of high-quality professional tennis match data, enriching it with derived metrics and several alternative rating formulations (including variants inspired by Elo- and Glicko-type ideas), and, finally, applying multiple machine learning approaches and comparing them under a common protocol, so that differences in predictive performance can be interpreted in light of consistent data and feature engineering.

# 3 Methodology

[You may structure this as you wish, to suit your particular project. Include information on the stages of the project and details of how each stage was carried out. In addition, explain and justify (either from common knowledge in the field, with reference to sources or through logical deduction) the choice of approach at all levels. A good rule of thumb for checking whether the argumentation in this (or any other) section is 'watertight' is to ask yourself if there is anything in the text that the reader could reasonably ask 'why' or 'how can you be sure?' about without getting an answer in the text itself – if there is, then you have more writing to do.

Examples of project stages are data collection and preparation, modelling, model selection and tuning, model evaluation and data interpretation (this is not an exhaustive list). However, do not include each and every type of task only because you learnt about in the course. Many will not be applicable in your project (e.g. data cleaning, scaling or PCA will not always be needed). If you are 'skipping' a common task such as data cleaning, just explain why, as this one of the decisions you made with regard to the methodology.

Discuss the implications of the methodology in the application domain.

Information on tools (e.g. Colab, Python libraries etc.) should be presented in a subsection of this chapter and not 'woven' through the conceptual account of the methodology.]

# 4 Results

[You may structure this as you wish, but let the reader know everything you know about the results, regardless of whether you consider the outcome good or bad.

Discuss the results, rather than simply presenting them to your reader and letting them do the work.

Be inquisitive when analysing your results. For example, if you are getting accuracies in the high 90s there is probably either an extreme imbalance in the data set or a data leak. Always investigate and at least discuss but, if there is time, it is best to redesign and redo your experiments.

Discuss the implication of the results in the application domain]

# 5 Conclusion

[Structure as you wish but make sure that a it includes a synthesis of the results, a high-level discussion of their implications in the application domain, any particularly interesting low-level details of the results and, finally, some words on future work i.e. how you or someone else might build on the project.]

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# A Examples of figure, table and equation

## Figure example

Example of figure:

Figure 1: Example caption

And here is an example of a reference to Figure 1.

## Table example

Example of table:

Table 1: Correlation values for N=100 and $\alpha = .05$

| Pair | r | Significant |
|---|---|---|
| First | 0.22 | Yes |
| Second | 0.07 | No |
| Third | 0.31 | Yes |

An example of referencing Table 1.

## Equation example

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{(n-1)S_x S_y} = \frac{S_{xy}^2}{S_x S_y} \tag{1}$$

where $x$, $y$ are the two variables, $x_i$, $y_i$ the i$^{\text{th}}$ values, $\bar{x}$, $\bar{y}$ the means, $S_x$ and $S_y$ the sample standard deviations and $S_{xy}^2$ the sample covariance. And here is a reference to Equation 1.]